The Asymmetrical Protocols Digital Signature

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- ## Definition of the Signature

A signature is a person's name written by him(her)self as a proof of authorship of the contents of a document.

- ## Peculiarities of the Signature

The signature should be authentic, unforgeable, non-reusable, unalterable and of be non-repudiable.

- And, this is how one can obtain the same characteristics digitally !!!

$\mathcal{H}=h(m)$

A hash function of *h* is given as H=h(m), where

h : Hash function,

m : variable length message,

H : fixed length hash value of the message.

Also known as:

- The message digest,
- The fingerprint or,
- The digital fingerprint of the message.

$\mathcal{H}=h(m)$

- The hash value (H) should be calculated relatively easily for any given message (m).
- h(m) should be a one-way function (computationally intractable).
- There should be only one H value for any given m (collision free).

$\mathcal{H}=h(m)$

- SHA-1 (Secure Hash Algorithm-1) was announced as a standard hashing algorithm FIPS PUB 180-1 by National Institute for Standards and Technology (NIST) of USA in 17.4.1995 and it produced 160 bit H value.
- NIST announced FIPS PUB 180-2 in August 1 2002. With this update, SHA-256, SHA-384 & SHA-512 surfaced.
- H values from SHA family are 160, 256, 384 & 512 bits respectively, in FIPS PUB 180-2.

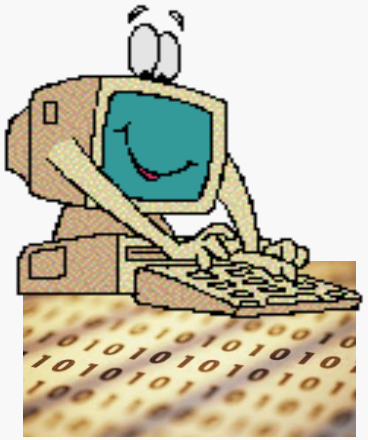
- MD5-SHA0 (160 bits)-SHA1 (256 bits) have already been broken thus must not be employed. SHA2 family (384 bits and 512 bits) should be used instead.

- Wang, X., Feng, D., Lai, X., Yu, H., ”Collisions for Hash Functions MD4, MD5, HAVAL-128 and RIPEMD”, Cryptology ePrint Archive, Report no: 2004/199, 2004.
- Wang, X., Yu, H., Yin, Y. L.,”Efficient Collision Search Attacks on SHA-0.”, Springer,Lecture Notes in Computer Science, v.3621, pp.1-16, 2005.
- Wang, X., Yin, Y.L., Yu, H., “Finding Collisions in the Full SHA-1”, Springer,Lecture Notes in Computer Science, v.3621, pp.17-36, 2005.

Digital signature

Buy the items and send them...

plaintext



Apply hash function

Secure Hash Algorithm (SHA)

Hash value

we09854jkl
)/(&hj37894

160 bits

Alice encrypts the hash value with her private (secret) key

+

Hash value

we09854jkl
)/(&hj37894

=

DIGITAL SIGNATURE

of Alice